

ProxSim: GPU-based Simulation Framework for Cross-Layer Approximate DNN Optimization

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Abstract—Through cross-layer approximation of Deep Neural Networks (DNN) significant improvements in hardware resources utilization for DNN applications can be achieved. This comes at the cost of accuracy degradation, which can be compensated through different optimization methods. However, DNN optimization is highly time-consuming in existing simulation frameworks for cross-layer DNN approximation, as they are usually implemented for CPU usage only. Specially for large-scale image processing tasks, the need of a more efficient simulation framework is evident. In this paper we present *ProxSim*, a specialized, GPU-accelerated simulation framework for approximate hardware, based on Tensorflow, which supports approximate DNN inference and retraining. Additionally, we propose a novel hardware-aware regularization technique for approximate DNN optimization. By using *ProxSim*, we report up to $11\times$ savings in execution time, compared to a multi-thread CPU-based framework, and an accuracy recovery of up to 30% for three case studies of image classification with MNIST, CIFAR-10 and ImageNet.

Index Terms—Simulation Framework, embedded Neural Networks, Approximate Computing, Deep Learning

I. INTRODUCTION

Deep Learning (DL) applications for image processing, such as Convolutional Neural Networks (CNNs), can achieve high, near-human accuracy at complex regression and classification tasks. This comes at the cost of large computational requirements, as DNNs perform energy-costly Multiply-and-Accumulate (MAC) operations and memory accesses. Through cross-layer approximate computing, significant savings in computational resources can be achieved by systematically introducing approximations in DNN computations at software and hardware levels. The introduced approximations cause accuracy degradation that may then reach critical levels, from which a recovery is possible only through optimization techniques. To analyze which approximation approaches are adequate for a certain DNN, and to perform the corresponding optimization for accuracy recovery, a specialized DL simulation framework for cross-layer approximation is required. Furthermore, the simulation of approximate, large-scale image processing tasks requires high computing parallelism, which can be achieved only through functional acceleration, for example by using Graphic Processing Units (GPU).

In this work, we present *ProxSim*, a specialized, highly flexible, GPU-accelerated Simulation Framework for testing and optimization of cross-layer DNN approximation. *ProxSim*

allows to retrain DNN parameters given specific approximate hardware, through a combination of existing techniques, and our novel regularization approach for DNN retraining. We summarize our main contributions as follows:

- *ProxSim*, a GPU-based, cross-layer approximate simulation framework for DNN optimization.
- Accelerated simulation of approximate hardware in DNN computation through specialized GPU support.
- Formulation of a novel hardware-aware regularization for approximate DNN retraining.
- Exploration and analysis of several techniques for optimization and retraining of approximate DNNs, achieving over 30% of accuracy recovery for the most extreme DNN approximations, compared to the initial accuracy.

We evaluate and validate our contributions through detailed experiments, focused on CNNs for image classification trained with MNIST [1], CIFAR-10 [2] and ImageNet [3].

II. RELATED WORK

Highly specialized, GPU-accelerated simulation frameworks for precision-scaling or quantization methods, such as Tensorflow [4], Graffitist [5] or Ristretto [6], have been proposed in the literature, regarding software-oriented approximation techniques. Simulation frameworks for cross-layer approximation of DNNs, on the other hand, are typically oriented towards specialized goals and thus are generally implemented for CPU only, which increments training and inference times, particularly when simulating approximate hardware on large DNNs. Examples of these frameworks are AxDNN [7] and TypeCNN [8]. AxDNN is a pre-RTL simulation framework for design space exploration which combines precision-scaling and pruning methods with simulation of approximate hardware. TypeCNN focuses on variable data types for accurate and approximate arithmetic operations for training and inference of small CNNs that are to run on dedicated hardware. While algorithmic simulation frameworks for cross-layer DNN approximations such as ApproxANN [9] and AxNN [10] have delivered efficient methods for partial DNN approximation, we focus on evaluating and optimizing full DNN approximation, to maximize energy savings. Recent works, such as AxTrain [11], have demonstrated that it is possible to recover lost accuracy in small, fully approximated DNNs through adapted Stochastic Gradient Descent (SGD) techniques. In